

Course on scRNA-seq Data Analysis



- Day 1 – Introduction and QC
- Day 2 – Data processing and visualisation
- Day 3 – Annotation and interpretations
- **Day 4 – Group work and presentations**

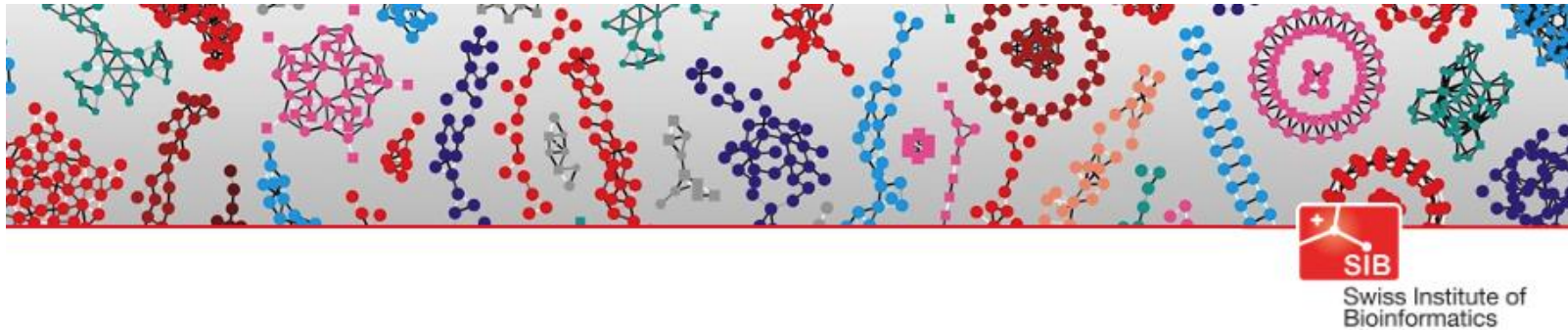


Swiss Institute of
Bioinformatics



Course on scRNA-seq Data Analysis

Based on the course by SIB <https://sib-swiss.github.io/single-cell-r-training/>



Single cell transcriptomics

Rachel Marcone Jeitziner

Tania Wyss

Alex Lederer

Geert van Geest



Swiss Institute of
Bioinformatics



Single-Cell Omics Community

Vibe coding

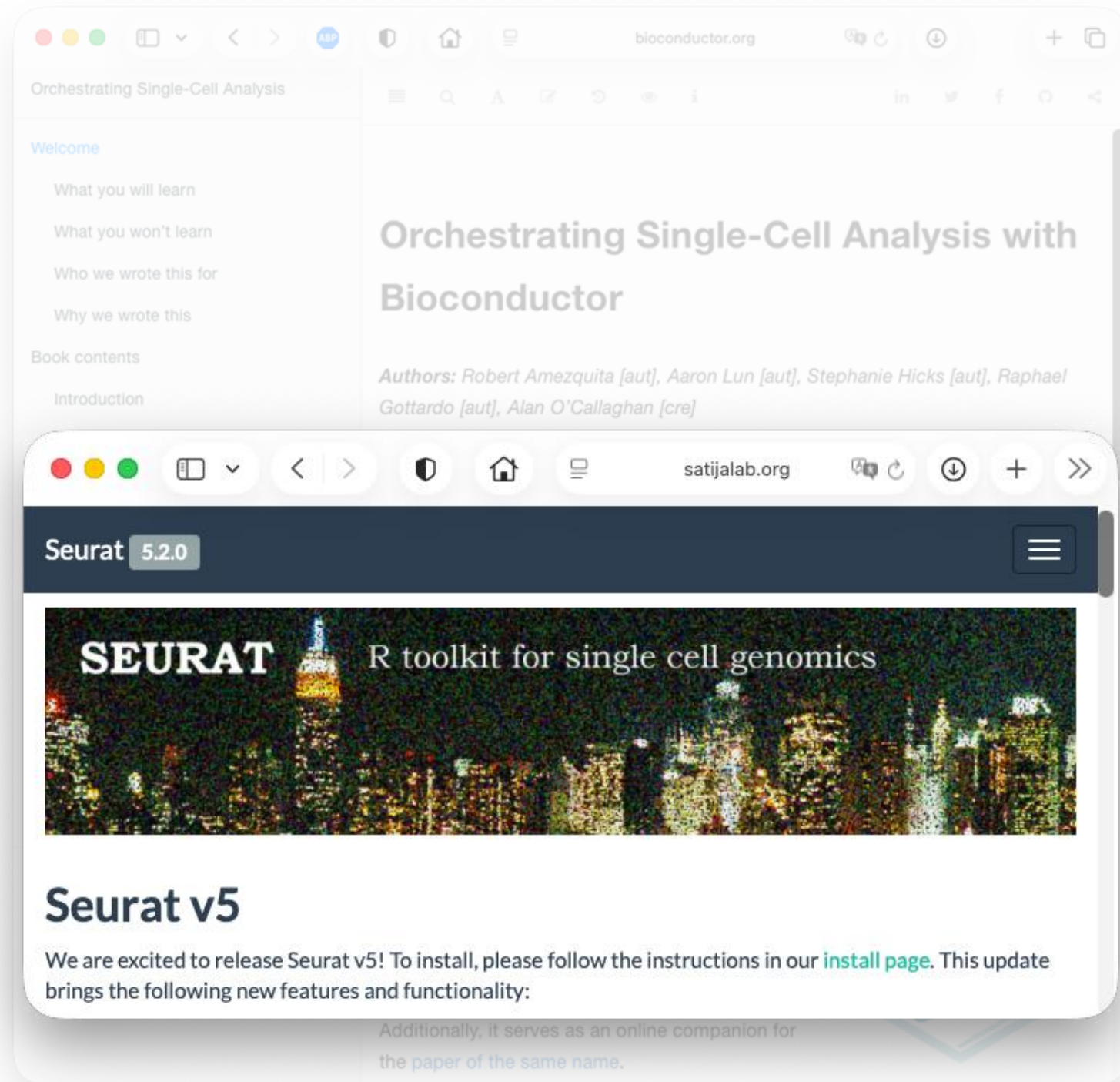
- OSCA book



A screenshot of a web browser displaying the landing page for the book "Orchestrating Single-Cell Analysis with Bioconductor". The browser's address bar shows "bioconductor.org". The page has a light blue header with the title "Orchestrating Single-Cell Analysis" and a navigation menu. The main content area is divided into two columns. The left column contains a "Welcome" section with links to "What you will learn", "What you won't learn", "Who we wrote this for", and "Why we wrote this". Below this is a "Book contents" section with links to "Introduction", "Basics", "Advanced", "Multi-sample", and "Workflows". The "Contributors" section lists "Aaron Lun, PhD", "Robert Amezquita, PhD", "Stephanie Hicks, PhD", and "Raphael Gottardo, PhD". The "Acknowledgements" section is also present. The right column features the book title "Orchestrating Single-Cell Analysis with Bioconductor" in a large, bold font. Below the title, the authors are listed: "Authors: Robert Amezquita [aut], Aaron Lun [aut], Stephanie Hicks [aut], Raphael Gottardo [aut], Alan O'Callaghan [cre]". The version is "Version: 1.20.0", the modification date is "Modified: 2022-09-05", the compilation date is "Compiled: 2025-11-03", the environment is "Environment: R version 4.5.1 Patched (2025-08-23 r88802), Bioconductor 3.22", the license is "License: CC BY 4.0", and the copyright is "Copyright: Bioconductor, 2020". The source is given as "Source: https://github.com/OSCA-source/OSCA". A "Welcome" section follows, stating: "This is the landing page for the 'Orchestrating Single-Cell Analysis with Bioconductor' book, which teaches users some common workflows for the analysis of single-cell RNA-seq data (scRNA-seq). This book will show you how to make use of cutting-edge Bioconductor tools to process, analyze, visualize, and explore scRNA-seq data. Additionally, it serves as an online companion for the paper of the same name." The Bioconductor logo is in the bottom right corner.

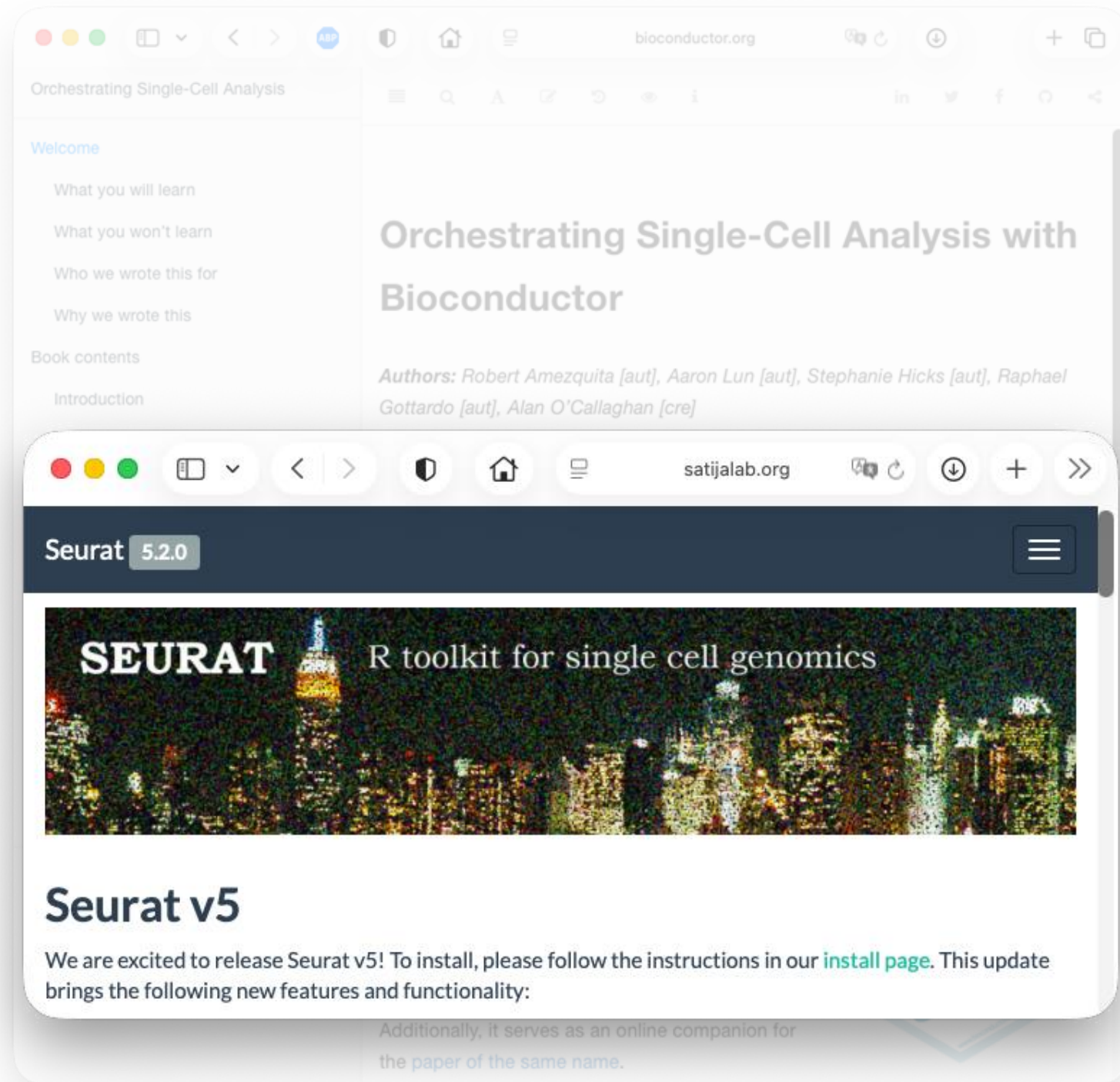
Vibe coding

- OSCA book
- Seurat documentation



Vibe coding

- OSCA book
- Seurat documentation
- Claude, Copilot and other LLMs
- RStudio and VS Code with LLM integrated



Serious coding

- OSCA book
- Seurat documentation
- Claude, Copilot and other LLMs
- RStudio and VS Code with LLM integrated
- scdrake
(our in-house pipeline)

scdrake 1.7.1

scdrake

NEWS updates Documentation & vignettes bioinfocfz.github.io/scdrake

Documentation & vignettes bioinfocfz.github.io/scdrake

Overview & outputs vignette("pipeline overview") Pipeline diagram Show

license MIT lifecycle stable Docker Image CI passing

[scdrake](#) is a scalable and reproducible pipeline for secondary analysis of droplet-based single-cell RNA-seq data (scRNA-seq) and spot-based spatial transcriptomics data (SRT). [scdrake](#) is an R package built on top of the [drake](#) package, a [Make](#)-like pipeline toolkit for [R language](#).

The main features of the [scdrake](#) pipeline are:

Links

[Browse source code](#)
[Report a bug](#)

License

[MIT](#) + file [LICENSE](#)

Community

[Contributing guide](#)
[Code of conduct](#)

Citation

[Citing scdrake](#)

Published by Bioconductor

Seurat v5

We are excited to release Seurat v5! To install, please follow the instructions in our [install page](#). This update brings the following new features and functionality:

which teaches users some common workflows for the analysis of single-cell RNA-seq data (scRNA-seq). This book will show you how to make use of cutting-edge Bioconductor tools to process, analyze, visualize, and explore scRNA-seq data. Additionally, it serves as an online companion for the paper of the same name.

Bioconductor

Course on scRNA-seq Data Analysis

Trainers

Jan Kubovčíak (IMG)

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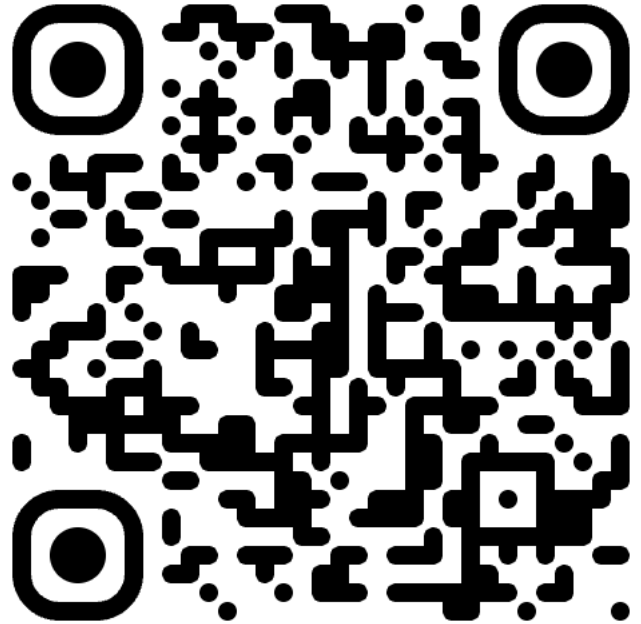
Organizators

Priit Adler

Diana Pilvar



Please provide your feedback





Tarantino et al., Pulp fiction, 1994